Nathan Pereira

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Skills

Engineering Softwares: Siemens NX, Solidworks, Fusion 360, AutoCad 2D/3D, Blender, Revit, Ardunio, Ansys
Technical Skills: CNC Machining, Additive Manufacturing, Project Management, GD&T, FEA, Root Cause Analysis

Education

University Of Waterloo — Waterloo, Ontario
Bachelor Of Applied Science, Mechanical Engineering — Graduation: April 2026

Work Experience

Dentra | *Co-Founder* — December 2024 - September 2025

- Achieved award by Conrad business school, receiving two paid interns for summer 2025, a total of **$10,000** investment, in addition winning **University of Waterloo's $500 Cornerstone award.**
- Successfully developed **Quantitative light fluorescence** and a rooted deep learning model to detect levels of plaque build up, being the first of its kind to be adaptable to everyday phones. Incorporating manufacturing methods of **silicone-based mold design** and **pcb design**.

Solid Ultrabattery Inc | *Battery Development Intern* — September 2024 - December 2024

- Researched and tested graphite anodes for cells, utilizing **thermogravimetric analysis** and **X-ray fluorescence testing** to determine material composition.
- Designed and tested a peristaltic pump and ventilation system for optimizing the cathode coating process, achieving **50%** reduction in production time, utilizing **SolidWorks** for design and **Ansys** for **CFD analysis**.
- Developed 3D-printed single and multi-layer pouch cell molds, streamlining the manufacturing process and achieving a **25%** reduction in production time.

Tesla | *R&D Process Engineering Intern - Cell Engineering* — January 2024 - April 2024

- Led the testing and validation of POP designs to improve **40 electrolyte filling machines** in Texas Gigafactory, utilizing in-depth procedures and root cause analysis to optimize designs.
- Spearheaded soak testing within a lab environment to evaluate the electrolyte compatibility of **18 material compositions** within cell manufacturing, applying a combination of qualitative and quantitative analysis.
- Conducted research and spearheaded the development of an RFID system slated for integration into upcoming next-generation electrolyte machines, representing an investment of over **$2.3 million**. Responsibilities encompassed rigorous wet lab material testing, design work, and validation of proof of concept.
- Conducted thorough **fatigue and dispense calibration cycling tests** on electrolyte pumps to pinpoint root causes and provide valuable feedback to designers. Utilized a **VHX microscope** for detailed analysis of degradation over time, enhancing our understanding of long-term performance.

Toyota | *Analyst Engineering Intern* — May 2023 - August 2023

- Utilized Toyotas' TBP problem analysis method to lead a cross-functional team in the complete redesign of the magnetic path for AGV navigation, optimizing route planning and eliminating potential disruptions, leading to improved overall operational reliability and a **62% decrease** in AGV errors.
- Engineered and implemented 3D-designed ergonomic hood jigs and fixtures, improving production efficiency and eliminating mechanical failures, resulting in **$5,000** in cost savings.

Cemcorp | *Mechanical Engineering Intern* — September 2022 - December 2022

- Prepared comprehensive process flow charts, **P&ID diagrams**, BOM, tie-point list, and equipment lists to aid seamless communication with stakeholders throughout the design and implementation phases of an industrial alcohol distillery plant.
- Designed 3D piping models for **stress analysis** and developed a comprehensive drawing package for an alcohol distillery plant, ensuring **ISO 9001 compliance**, which led to a **Co-op Student of the Year nomination**.

MCW Consultants | *Mechanical Engineering Intern* — January 2022 - April 2022

- Analyzed mechanical blueprints and markups, providing drafting of HVAC and Plumbing in **AutoCad** and **Revit** 3-D modelling for 10 different clients

Projects/Design Teams

Battery Workforce Challenge - *Solidworks, Ansys*
Designed an **EV battery module** using Solidworks, conducting research on structural materials, performing load simulations and ensuring all mechanical and structural standards are met for the competition.

UWMedTechResolve - *Solidworks, Project Management*
One of the team leads for the **conceptual design and development phase** of the University of Waterloos Medtech design team for the upcoming Form and Function competition. Performing **design sprints**, **R&D** and ensuring deliverables are met while leading **15 members**.